

Friction

Normal reaction and limiting equilibrium

YOUR AIM TODAY

Draw friction correctly and use $F \leq \mu R$.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

The ideas you need

Friction acts along a contact surface and opposes actual or impending motion.

- Normal reaction R acts perpendicular to the surface.
- Friction F acts parallel to the surface.
- Always $F \leq \mu R$; equality holds only at limiting equilibrium.
- On a horizontal surface with no other vertical forces, $R = mg$.
- Decide the likely direction of motion before drawing friction.

MEMORY RULE

Friction is not automatically μR ; use $F = \mu R$ only when limiting or about to slip.

Horizontal pull

WORKED STEPS

Mass 5 kg, $\mu = 0.3$, limiting: $R = 5g$, so $F = 0.3(5g) = 1.5g$ N.

Equilibrium

WORKED STEPS

If a horizontal pull P is balanced before limiting, $F = P$ and verify $P \leq \mu R$.

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. A 4 kg block rests horizontally. Find R.	
2. If $\mu=0.25$, find limiting friction.	
3. A 6 N pull keeps it at rest. Find actual friction.	
4. Will a 12 N pull move it? Use $g=9.8$.	
5. A 10 kg block is limiting under a horizontal force P; $\mu=0.4$. Find P.	
6. Explain why writing $F=\mu R$ for every problem is wrong.	

STOP

Do not continue until all six questions have an attempt.

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	$R=4g=39.2 \text{ N}$.
2	$F_{\max}=0.25(39.2)=9.8 \text{ N}$.
3	$F=6 \text{ N}$, opposite the pull.
4	Yes; $12 > 9.8 \text{ N}$.
5	$R=98 \text{ N}$; $P=F=0.4(98)=39.2 \text{ N}$.
6	Static friction adjusts up to its maximum; equality occurs only at limiting equilibrium.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.